

FOREST RESOURCES

3.1

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A forest is a community of trees, bushes, and other woody vegetation in a closed atmosphere. India has a forest cover of 6,76,000 sq.km (20.55% of the total geographic area). Scientists suggest that India should have 33% of its land covered in forests. Today, we only have around 12%, therefore we need not just to protect our present forests while also increasing our forest cover.

3.1.1 Uses of Forest

1. Commercial uses

- ★ Wood – used as a fuel
- ★ Supply wood for various industries – Raw materials as pulp, paper, furniture timber etc.
- ★ Minor forest products – gum, dyes, resins
- ★ Many plants – Medicines
- ★ Supply variety of animal products – honey. Ivory, horns etc.
- ★ Many forest lands are used for - Mining, grazing, for dams and recreation.

2. Ecological uses

1. **Production of oxygen:** Photosynthesis generates significant amounts of oxygen, which is necessary for living.
2. **Reducing global warming:** Carbon dioxide is one of the primary greenhouse gases. Plants absorb it for photosynthesis. As a result, the problem of CO₂-induced global warming is decreased.

3. **Soil conservation:** Tree roots tie the soil together tightly, preventing erosion. They also serve as windbreaks.
4. **Regulation of hydrological cycle:** Watersheds in forests act like huge sponges, slowly releasing water to recharge springs.
5. **Pollution moderators:** Forests absorb harmful gases and noises, preventing pollution.
6. **Wild life habitat:** The forest is home to millions of wild creatures and vegetation.

3.1.2 Types of Forest

There are six types of forest in india

1. Tropical evergreen forests.
2. The tropical deciduous forests.
3. Scrub forests and grassland.
4. Desert vegetation.
5. Mountain forests and
6. Mangrove forests.

1. Tropical evergreen forests

Tropical evergreen forests do not shed their leaves all at once, thus they remain green throughout the year. These forests are found in locations with high temperatures and rainfall that exceeds 250 cm.

Place: Assam, Nagaland, Meghalaya.

Trees: Teak, Rosewood, Ebony.

2. The tropical deciduous forests

Tropical deciduous forests grow in areas with substantial rainfall for a portion of the year, followed by a significant dry season.

Place: Maharashtra, Karnataka, Kerala, and Tamil Nadu.

Trees: Teak Sal, Sandalwood, Kusum.

3. Scrub forests and grassland

This vegetation is composed of many southern tropical thorns, which form scrub-type forests. These forests are typically found in Kutch and on the outskirts of the Thar desert. Rainfall in these forests ranges between 60 and 100 cm.

Place: eastern Rajasthan, Punjab.

Trees: coarse grass, prickly trees, and bushes.

4. Desert vegetation

The vegetation in the desert is dry and sparse, with rainfall ranging from 10 to 50 cm. These deserts are made of sand that has collected over the last 1.8 million years.

Place: Punjab, Haryana, Rajasthan

Trees: Cacti, Acacias, Palms

5. Mountain forests

Mountain forests can be found on the slopes of mountains, particularly in the Himalayas and, to a lesser extent, in the Niligiri Hills.

Trees: oak, chestnut, ash, beech, pine.

6. Mangrove forests

Mangroves are a unique form of vegetation. And they can be found in inter tidal areas where freshwater and saltwater combine, such as bays, estuaries, creeks, and lagoons.

3.1.3 Over-Exploitation of Forests

Causes of Over-Exploitation of Forests

- ★ Deforestation for Agriculture
- ★ Overharvesting of Timber and Wood
- ★ Mining and Extractive Industries
- ★ Forest Fires (Both Natural and Man-Made)
- ★ Overgrazing by Livestock
- ★ Industrialization and Urban Expansion.

Consequences of Over-Exploitation of Forests

- ★ Loss of Biodiversity
- ★ Climate Change Acceleration
- ★ Soil Erosion and Desertification
- ★ Displacement of Indigenous Communities
- ★ Water Cycle Disruption
- ★ Economic Losses and Reduced Livelihoods

3.2**DEFORESTATION**

Deforestation refers to the elimination of forest resources through natural or man-made processes.

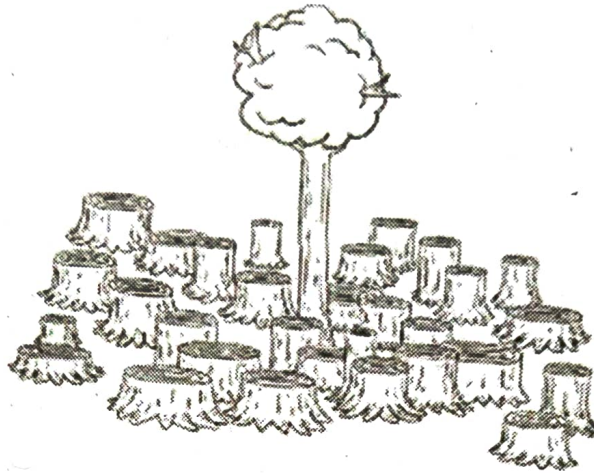


Figure 3.1 Deforestation

3.2.1 Causes of Deforestation

1. **Logging** : Illegal logging activities are common, destroying the livelihoods of those who rely on forests.
2. **Agricultural Activities** : The conversion of forests to agricultural land is a major cause of deforestation. Many trees are cut down to meet the increased need for food goods. More than 40% of the forests are cleared to make way for agriculture and timber production.
3. **Mining** : Oil and coal extraction both require a significant amount of forest area. Road construction causes deforestation because it allows access to remote areas. Mining waste pollutes the environment and has an impact on local wildlife.
4. **Urbanization** : People's needs increase as the population grows, contributing to deforestation. Forests decrease significantly to accommodate demands such as road construction, housing development, mineral exploitation, and industrial expansion.
5. **Timber Production** : Timber industry is a main cause of deforestation. There is a high demand for timber, which leads to increased deforestation. It is a raw material used in both paper manufacture and construction.

6. **Forest Fires** : Every year, we lose a great number of trees to forest fires in various parts of the world. This is caused by severe summer and winter temperatures. Fires, whether caused by humans or nature, result in massive losses of forest cover.

3.2.2 Effects on Deforestation

1. **Green Gas Emission** : Gases like methane and carbon dioxide trap heat in the earth's atmosphere, leading to climate change.
2. **Soil Erosion** : Cutting down trees leads to forest clearance, which causes soil erosion.
3. **Bio diversity losses** : Deforestation affects land, making it difficult for plants and animals to live. This can lead to the extinction of entire species due to flooding, resulting in bio diversity losses.
4. **Expansion of Deserts** : Due to the acts of powerful winds, large areas of land are gradually covered by sand deserts.
5. **Decrease in Rainfall** : In the absence of forest, rainfall reduces significantly because forests provide rainfall and maintain high humidity in the atmosphere.
6. **Loss of fertile land** : Less rainfall leads to the loss of fertile land since there is less natural vegetation growth.

3.2.3 Prevention Measure

1. Role of Individuals

Implementation of the three R's (Reduce, Reuse, and Recycle).

1. **Reduce**: Reducing paper use and replacing alternatives.
2. **Reuse**: Avoiding products that are only intended to be used once and then destroyed in order to reduce waste.
3. **Recycle**: Recycle all paper and wood products properly.
4. Spreading awareness about the value of trees and the adverse effects of deforestation.
5. Planting more and more trees to compensate for tree destruction, which will ultimately benefit humanity.

2. Role of Government

1. Launching new forestry projects to recover destroyed lands.

2. Investing in advanced farming technologies like hydroponics.
3. Supporting farmers in implementing environmentally friendly agricultural practices, such as cycle agriculture.
4. Strict measures are being implemented to prevent unlawful forestry activity.
5. To decrease forest loss, carefully implement construction plans for infrastructure such as roads, industries, residences, dams, and so on.

3.2.4 Case Study

The deforestation in the Andaman Islands

Reasons:

Following India's 1947 independence from the British, thousands of people were sent to the islands as part of a colonization plan, which raised the need for resources.

The wood found in the many forests served as the foundation for the creation of jobs and income.

The Andaman Islands' timber-based economy currently consists of two government sawmills, a few small commercial sawmills and furniture manufacturing companies, and three private plywood companies.

3.3

MINING

Mining operations for minerals and fossil fuels, such as coal, sometimes require extensive forest tracts. Surface mining is used for shallow deposits, while deep deposits are mined sub-surface.

More than 80,000 hectares of land in the country are now under the stress of mining operations. Mining activities need the removal of plants, soil, and rock. This defaces the topography and destroys the landscape in the area.

Deforestation has been recorded in Mussorie and Dehradun valleys due to uncontrolled mineral mining over a 40-kilometer stretch.

3.3.1 Effects or impacts of Mining

Water impacts

- ★ Changes in hydrology

- ★ Ground and surface water impacts
- ★ Changes in water quality, clouding of waterways, indirect impacts on plants and animals
- ★ Controlled wastewater discharges and their cumulative impacts
- ★ Increasing water scarcity in arid areas
- ★ Risks of contamination, acid mine drainage, dam accidents, dewatering.

Environmental impacts :

- ★ Climate impacts: carbon dioxide and other greenhouse gases
- ★ Air quality and dust
- ★ Noise
- ★ Land vibration
- ★ Light
- ★ Smell
- ★ Radiation
- ★ Cumulative impacts

Economic and employment impacts:

- ★ Downstream impacts, enabling other industries
- ★ Tax income, royalties
- ★ Employment and contracting opportunities
- ★ Opportunities for new businesses, investments
- ★ Impact on existing livelihoods
- ★ Economic risks: bankruptcy, increased dependency

Land impacts:

- ★ Waste rock and tailings
- ★ Habitat changes, habitat fragmentation, and habitat loss
- ★ Loss of biodiversity and vegetation, including deforestation
- ★ Erosion
- ★ Alteration of soil profiles and risks of contamination.

3.4**DAMS ON FOREST**

Over 45,000 major dams worldwide contribute to economic development by harnessing water resources. According to current estimates, dams account for 30-40% of all irrigated land worldwide. Hydropower, Stored water accounts for 19% of global electricity supply and is used in over 150 nations. China and India, the world's two most populous countries, have built around 57% of major dams worldwide.

3.4.1 Dam Scenario

Dams	States	Gross storage capacity
Tehri	Uttarkhand	3540
Sardar sarover	Gujarat	9500
sholayar	Tamil Nadu	152.48
Bhakra	Himachal Pradesh	9867.84
Aruna	Maharashtra	93.378

3.4.2 Effects of dams on forest

1. Thousands of hectares of forest will be cleared.
2. Killing wild animals and destroying aquatic life.
3. Spread of waterborne illnesses.
4. Waterlogging raises the pH level of the soil.
5. In recent years, hydroelectric projects have caused widespread forest loss.

Example: The Narmada Sagar project has submerged 3.5 lakh hectares of forests.

3.4.3 Effects of Dam on Tribal People

- ★ The construction of large dams causes the displacement of tribal people.
- ★ Displacement and cultural change have an impact on traditional people's minds and bodies.
- ★ They do not accommodate modern eating patterns and lifestyles.
- ★ Tribal people are mistreated in modern society.

- ★ Many displaced people were not recognized, resettled, or paid.
- ★ Tribal people's bodies may not adapt to new environments, leading to increased disease risk.

3.5**TIMBER EXTRACTION****Definitions**

Timber extraction is the process of cutting and removing trees from forests for commercial use such as furniture, construction, and paper.

3.5.1 Types of Timber Extraction

- (i) **Clear felling** : All trees in a specific area are cut down regardless of species or size.
- (ii) **Selective logging** : In this logging, only large individual trees of a few economically marketable species are harvested.
- (iii) **Mechanized logging** : In this logging, heavy machineries are used to pull, lift and transport the trees. This process can be used in clear felling or selective logging operations.
- (iv) **Hand logging** : It takes place in forests that are seasonally flooded or permanently water logged. In such conditions, heavy machinery cannot be used and commercial logging operations have to employ hand-logging.
- (v) **Reduced-impact logging** : It is now a common feature in industrialized nations where environmental damage can be minimized through the selection of site-sensitive techniques of harvesting and logging.

3.5.2 Uses of timber

1. Timber is a valuable forest resource used for construction, furniture, and artifacts.
2. It is utilized in the production of boats, carts, carriages, and agricultural equipment.

3.5.3 Effects of Timber Extraction

The major effects of Timber extraction on forest and tribal people include :

- ★ Poor logging results in a degraded forest.
- ★ Floods may be intensified by cutting of trees or upstream watersheds.
- ★ Loss of biodiversity.
- ★ Climatic changes such as lower precipitation.
- ★ New logging roads permit shifting cultivators to gain access to logged areas and fell the remaining trees.
- ★ It results in forest fragmentation which promotes loss of biodiversity because some species of plants and animals require large continuous areas of similar habitat to survive.
- ★ Exploitation of tribal people by the contractors.
- ★ Soil erosion specially on slopes occurs extensively.
- ★ Sedimentation of irrigation systems, floods may be intensified by cutting of trees on upstream.